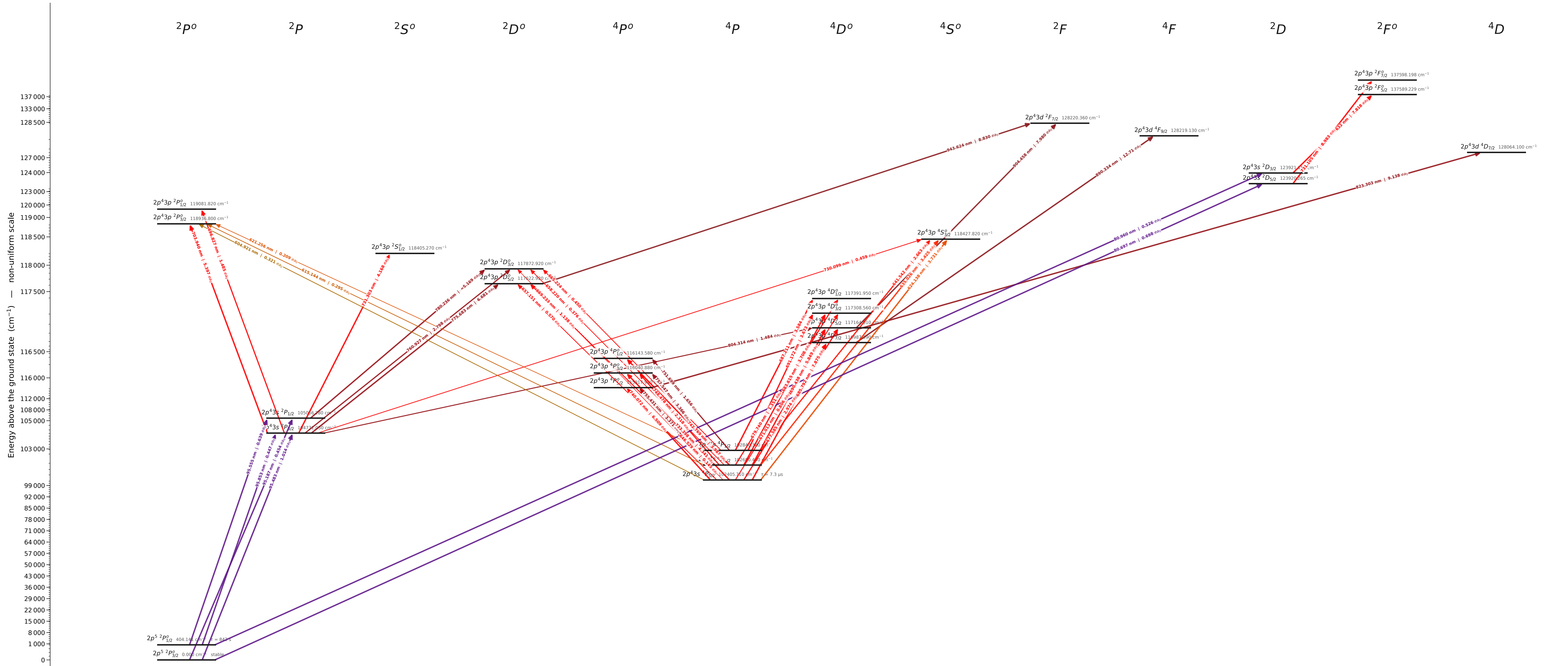
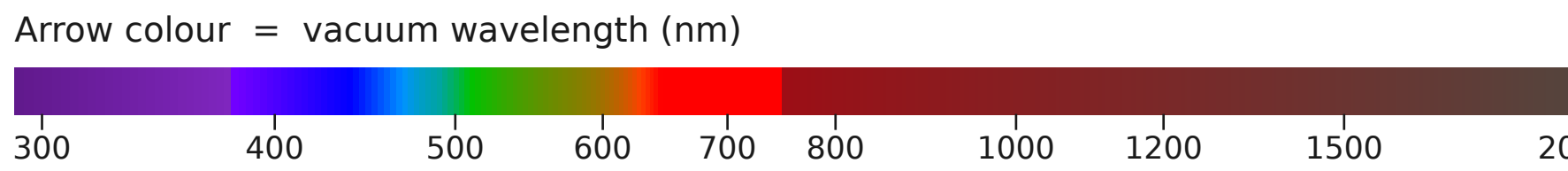


Fluorine

F (F I). Strongest classified transitions below 2 μm (NIST): 27 levels, 46 lines.

F⁺ ionisation limit 140524.50 cm⁻¹ = 17.42282 eV (71.162 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle f || er || i \rangle |$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e J' \rangle $ (ea)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$2p^3s^1\ ^4P_{5/2} - 2p^4s^1\ ^4D^o_{5/2}$	677.5848	677.3978	442.4427	2.974	9.600e+06	-	-	677.4 nm
$2p^3s^1\ ^4P_{5/2} - 2p^4s^1\ ^4D^o_{3/2}$	685.7918	685.6027	437.14789(15)	7.875	4.870e+07	-	-	685.6 nm
$2p^3s^1\ ^4P_{3/2} - 2p^4s^1\ ^4D^o_{3/2}$	760.9267	760.7173	393.9833	2.798	9.000e+06	-	-	761 nm: diode-laser absorption hyperfine spectroscopy
$2p^3s^1\ ^2P_{3/2} - 2p^4s^1\ ^2D^o_{3/2}$	775.6827	775.4692	386.4885	6.681	3.230e+07	-	-	775 nm: diode-laser saturated-absorption hyperfine spectr...
$2p^3s^1\ ^4P_{1/2} - 2p^4s^1\ ^4D^o_{1/2}$	780.2357	780.0211	384.2332	≈ 5.169	2.849e+07	-	-	780 nm: diode-laser Doppler-free saturated-absorption hyp...

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

^{19}F hyperfine structure ($I = 1/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$2p^2\ ^2P_{3/2}$	2009.990(10)	-	F = 1: -2512.49 F = 2: +1507.49
$2p^2\ ^2P_{1/2}$	10244.21(3)	-	F = 0: -7683.16 F = 1: +2561.05
$2p^4\ ^3s\ ^2P_{3/2}$	2643.0(10)	-	F = 2: -4625.25 F = 3: +3303.75
$2p^4\ ^3s\ ^2P_{3/2}$	3057.9(21)	-	F = 1: -3822.38 F = 2: +2293.43
$2p^4\ ^3s\ ^2P_{1/2}$	1737.1(14)	-	F = 0: -1302.83 F = 1: +434.27
$2p^4\ ^3p\ ^4P_{9/2}$	1824(20)	-	F = 1: -2280.00 F = 2: +1368.00
$2p^4\ ^3p\ ^4D_{9/2}$	1564.0(10)	-	F = 3: -3519.00 F = 4: +2737.00
$2p^4\ ^3p\ ^4D_{5/2}$	1148.0(10)	-	F = 2: -2009.00 F = 3: +1435.00
$2p^4\ ^3p\ ^2D_{5/2}$	1746.5(15)	-	F = 2: -3056.38 F = 3: +2183.12
$2p^4\ ^3p\ ^2D_{3/2}$	1857.1(21)	-	F = 1: -2321.38 F = 2: +1392.83
$2p^4\ ^3d\ ^4D_{1/2}$	793(20)	-	F = 3: -1784.25 F = 4: +1387.75

Measured A, B constants; sources in the data file